Experiment 02 — State-Replay Hazard Audit

Two-Layer CTMC reproducibility audit

Final report: 100 replications

Abstract

This experiment measures pure formula-level rate-abstraction error by evaluating exact active-edge and aggregate prevalence hazards on identical Full Micro states. This document reports 100 replay replications.

1 Research question and relation to Experiment 01

Experiment 01 separated hazard and placement mechanisms using separate stochastic trajectories. Experiment 02 removes trajectory, placement, and synchronization differences: it asks how much the aggregate formula differs from the exact inter-community transmission rate on the same microscopic Full Micro state.

2 Experimental design

The baseline is a chain of $L = 10$ connected sparse random communities, $n = 100$, mean internal degree approximately 6, and five unit-weight inter-community edges per adjacent pair. Rates are $\beta_{\text{in}} = \beta_{\text{out}} = \beta_{\text{macro}} = 0.003$, $\gamma = 0$, and $T_{\text{end}} = 4000$. No aggregate trajectory is simulated. The current stage contains 100 independent network/event replications; inference first aggregates within replication.

3 Mathematical definitions

For directed interface $i \to j$ and microscopic state x,

$$a^*_{ij}(x) = \beta_{\text{out}} \sum_{(u,v)\in E_{ij}} w_{uv} \mathbf{1}[X_u = I, X_v = S], \tag{1}$$

$$\lambda_{ij}(x) = \beta_{\text{macro}} W_{ij} \bar{i}_i(x) \bar{s}_j(x). \tag{2}$$

Signed, absolute, and symmetric relative errors are $\lambda - a^*$, $|\lambda - a^*|$, and $2|\lambda - a^*|/(\lambda + a^*)$ when the denominator is positive. All hazard-mass quantities are exact sums of piecewise-constant interval values times interval duration.

4 Reconstruction and replay

Full Micro was *not rerun*. The canonical Experiment 01 final-100 event archive supplies replications 1–100. With $\gamma = 0$, the seed and every node-level infection event determine the microscopic state

exactly. Networks are regenerated with the archived network seeds, and every event edge and community identifier is checked. A post-event state is used on $[t_r, t_{r+1})$ and the final interval is clipped at T_{end}. 35 logged events beyond the horizon were ignored before state mutation. Experiment 01 did not archive edge lists, so regeneration depends on the current network generator and NetworkX version; Experiment 02 records stable graph hashes.

5 State categories and stratification

The mutually exclusive categories are both-zero, premature-positive, missed-positive, overestimation, underestimation, and approximate-match. Approximate match uses absolute tolerance 10^{-12} and relative tolerance 10^{-9} after zero cases are classified. Premature-positive cases are retained explicitly. For monotone SI dynamics, source phases are not-started ($I/N = 0$), early ($0 < I/N \leq 0.10$), middle ($0.10 < I/N \leq 0.50$), and late ($I/N > 0.50$). These fixed bands distinguish establishment, growth, and saturation while remaining comparable across replications. The archived descriptive stratified table covers source infected and destination susceptible fractions, both boundary-node counts, active-edge count and weight, layer, both times since first local infection, and phase. Those interval-level summaries are descriptive only; layer and phase uncertainty is computed after within-replication aggregation.

6 Sanity and semantic checks

Automated tests cover brute-force exact hazards, the production aggregate formula, zero and positive active-edge cases, exact piecewise integration, premature mass, deterministic replay, reconstruction correctness, Experiment 01 identifiers, and production-file hashes. During replay, every state is also checked against the production MicroEngine total weighted infection rate and every aggregate entry against MacroEngine. Maximum replay differences were 5.7732e-15 (micro) and 3.4694e-18 (macro). 2026-07-17 final verification test run Command: .venv/bin/python -m unittest -v tests.test_experiment_02_state_replay_hazard_audit tests.test_hazard_placement_ablation Result: PASS Tests run: 18 (10 Experiment 02 audit tests plus 8 Experiment 01 regression tests) Failures: 0 Errors: 0 Coverage: exact hazard/brute force; aggregate/production formula; zero exact hazard; positive exact hazard; piecewise-constant integration; premature hazard mass; deterministic reproducibility; reconstruction and horizon correctness; Experiment 01 identifiers; production-module hashes.

7 Final results

The replay contains 99519 exact state intervals and 1791342 directed-interface rows. Confidence intervals below bootstrap replications, not event intervals.

Table 1: Replication-level final summaries; bootstrap 95% confidence intervals for the mean.

Metric	Mean	Median	95% CI
A_{exact}	27.131	27.106	[26.322, 27.93]
$A_{\text{aggregate}}$	46.078	46.208	[45.178, 47.023]
Signed hazard mass	18.947	18.684	[18.349, 19.56]
Absolute hazard mass	25.154	25.127	[24.653, 25.662]
Over-hazard mass	22.05	22.054	[21.545, 22.563]
Under-hazard mass	3.1035	3.0237	[2.8825, 3.3214]
Premature hazard mass	10.239	10.061	[9.8903, 10.602]
Missed hazard mass	0	0	[0, 0]
Premature fraction	0.22537	0.22269	[0.2156, 0.23611]

Table 2: Mean replication-level results by chain interface. Each interface includes both directions.

Interface	Signed mass	Absolute mass	Premature fraction
0	2.2416	2.8834	0.25964
1	2.3818	2.9031	0.24274
2	2.0465	2.7402	0.23522
3	1.9909	2.8122	0.25777
4	2.079	2.7037	0.23787
5	2.0818	2.6948	0.26631
6	2.1153	2.9129	0.25281
7	2.0739	2.8717	0.26972
8	1.9361	2.6318	0.23385

Table 3: Mean replication-level results by source epidemic phase.

Phase	Signed mass	Absolute mass	Premature fraction
early	0.071158	1.5434	0.79201
late	17.683	19.987	0.20022
middle	1.1928	3.6238	0.30165
not_started	0	0	NA

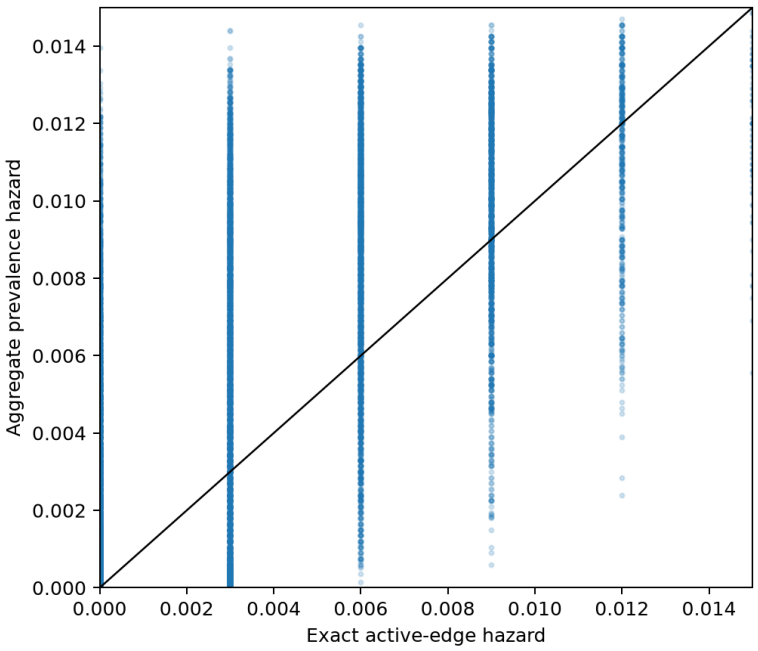


Figure 1: Exact and aggregate hazards evaluated on common Full Micro states. Points are a visualization subsample; inference is replication-level.

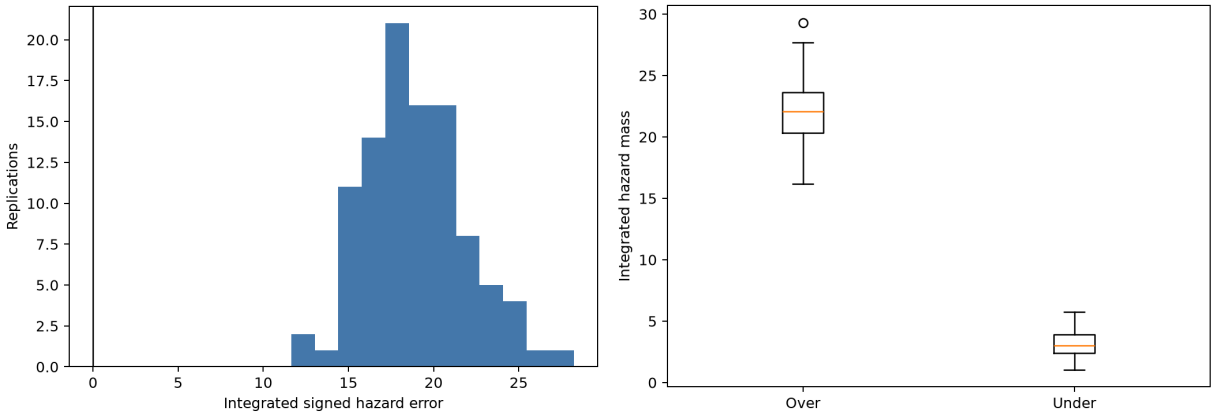


Figure 2: Replication distributions of integrated signed error and its over/under components.

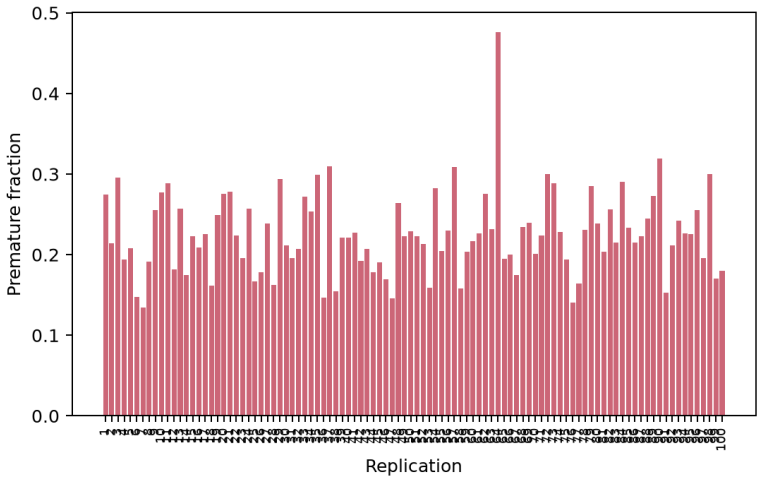


Figure 3: Fraction of aggregate cumulative hazard accumulated when exact inter-community transmission is impossible.

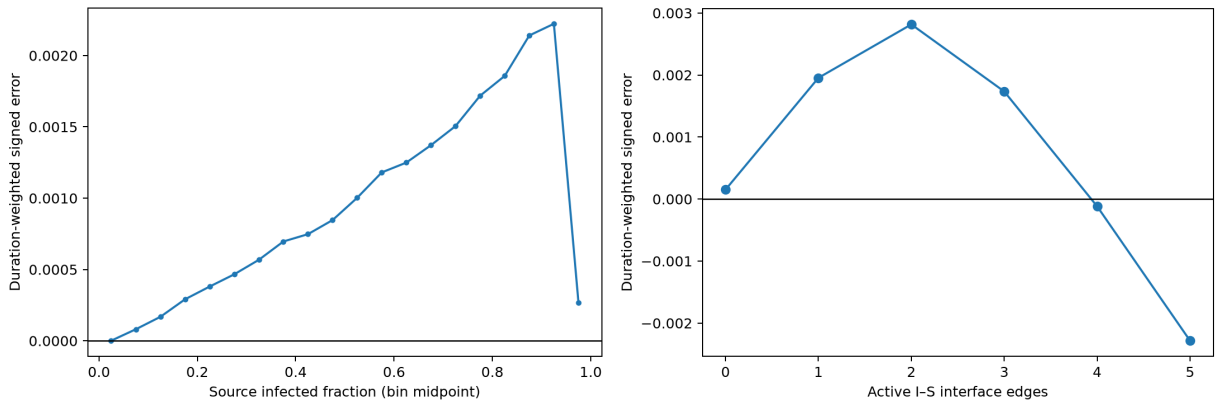


Figure 4: Descriptive event-level stratification. Intervals are dependent and are not treated as independent observations.

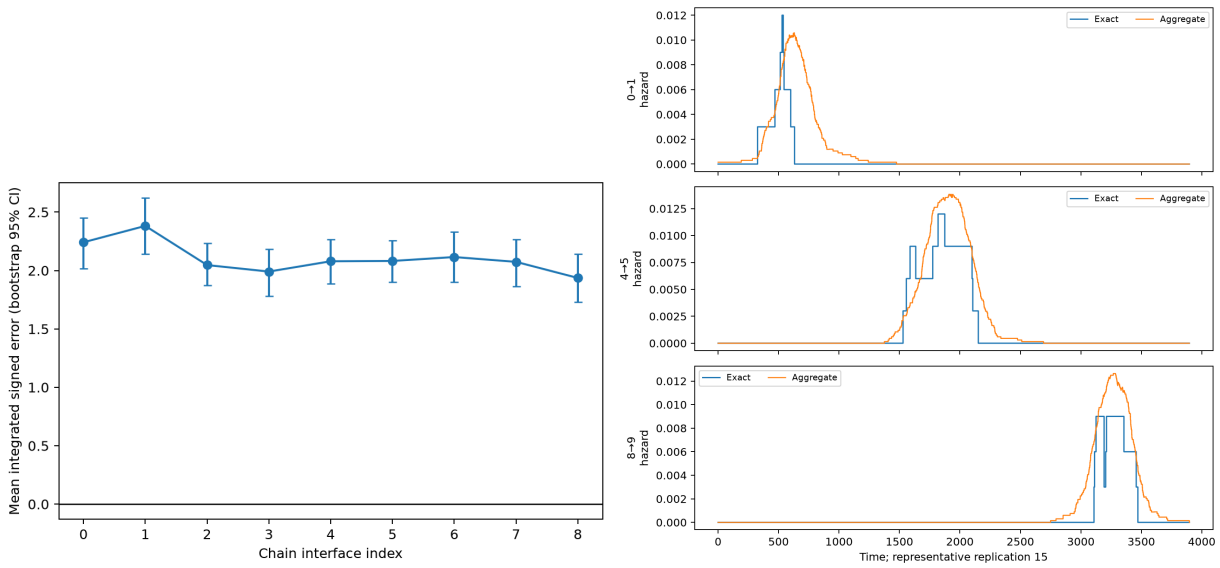


Figure 5: Layer-wise integrated error and representative common-state time series. Uncertainty claims use replication summaries.

8 Scientific interpretation

The signed mass must be interpreted together with absolute, over-, under-, and cumulative premature components: cancellation can make signed error small even when local discrepancies are large. The reported final estimates characterize pure formula-level rate-abstraction error for this baseline. Both over- and under-hazard mass are present, although the replication-level mean over-hazard component is larger for this baseline. This does not establish that aggregate hazards universally overestimate exact hazards.

Across replications, mean cumulative exact and aggregate hazards were 27.131 and 46.078; mean signed and absolute discrepancy masses were 18.947 and 25.154. Mean over-hazard mass (22.05) exceeded mean under-hazard mass (3.1035), but the nonzero under component confirms state-dependent sign changes. A mean 22.537% of aggregate cumulative hazard accumulated in states with zero exact active-edge hazard (95% bootstrap interval 21.56–23.611%). The phase-stratified mean premature fractions were 79.201% early, 30.165% middle, and 20.022% late. These phase values describe where formula mismatch occurs on the replayed trajectories; they are not independent event-level estimates.

9 What this experiment explains—and does not explain

It explains discrepancies between two hazard formulas on common Full Micro states. It cannot explain import-placement error, production synchronization error, full endogenous trajectory error, event-time bias by signed error alone, or robustness to other topologies or $\gamma > 0$.

10 Limitations

This final analysis has 100 replications. Network edges were regenerated rather than archived by Experiment 01. Event-level correlations are descriptive because intervals within a replication are

dependent. The phase thresholds are mechanistic fixed bands, not data-selected changepoints. Generalization remains limited to the chain topology and $\gamma = 0$ baseline.

11 Recommended next experiments

Experiment 03 is the Production vs Event-Driven Synchronization Audit, intended to explain the difference between approximately 7.5% error for event-driven aggregate/uniform Experiment 01 and approximately 42.6% in the production sparse benchmark. Experiment 04 is the Gateway-Distance Experiment with near/medium/far gateway placement at fixed n, m, d, k, β, and community count. Experiment 04 was not run here.

12 Exact reproduction commands

```
.venv/bin/python -m unittest tests.test_experiment_02_state_replay_hazard_audit
.venv/bin/python extrascripts/run_experiment_02_state_replay_hazard_audit.py --stage smoke --res
.venv/bin/python extrascripts/run_experiment_02_state_replay_hazard_audit.py --stage validation
.venv/bin/python extrascripts/analyze_experiment_02_state_replay_hazard_audit.py --stage validat
# Only after explicit approval:
.venv/bin/python extrascripts/run_experiment_02_state_replay_hazard_audit.py --stage final --aud
```